



MRC  
Biostatistics  
Unit



UNIVERSITY OF  
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# **Bayesian methods for clinical trials**

## *Lecture 3: Basket trial design*

Thomas Jaki & Pavel Mozgunov

MRC Biostatistics Unit

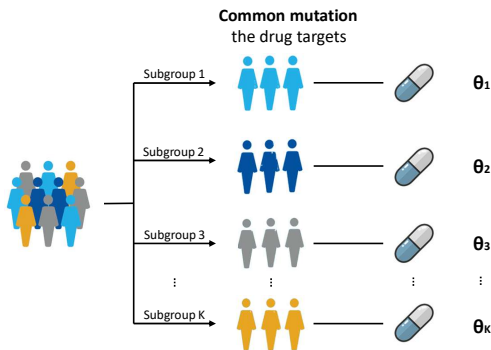
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# Basket trials

**Setting:** Common characteristic (e.g. mutation) present in multiple tumour types.

**Aim:** To develop **targeted therapies**

**Solution** Using biomarker(s) to screen patients and recruit those harbouring a common characteristic (mutation)

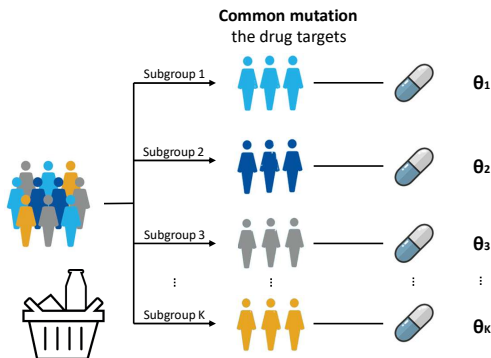


# Basket trials

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**Aim:** To develop **targeted therapies**

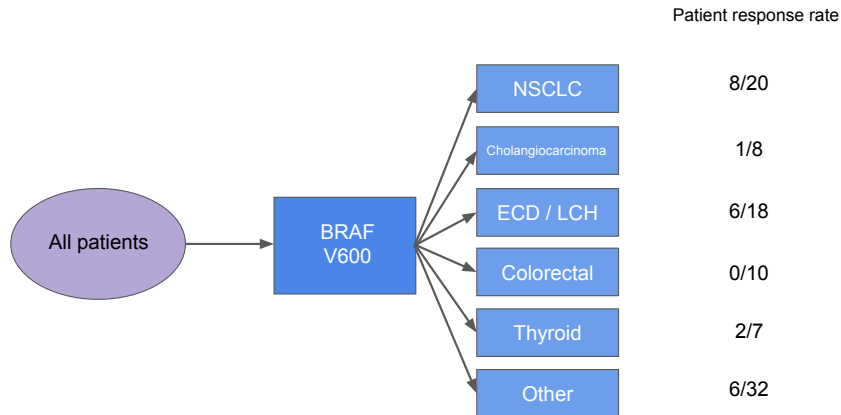
**Solution** Using biomarker(s) to screen patients and recruit those harbouring a common characteristic (mutation)



# Basket trials in oncology – An example

Hyman *et al.* (2015) reported a recent basket trial, which has been designed to evaluate the efficacy of vemurafenib in BRAF-V600.

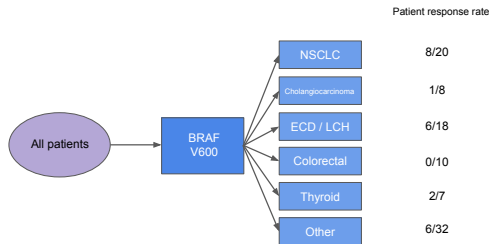
A total of 122 patients with BRAF-V600 mutations were enrolled, of which 95 entered the 6 modules.



# Borrowing of information between modules

With the **common genomic mutation** targeted by the investigational drug, one may expect:

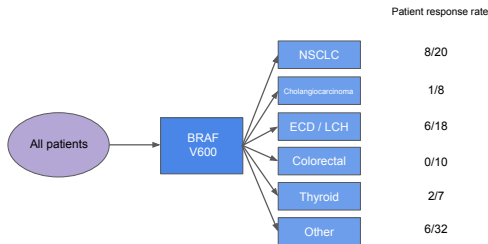
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# Borrowing of information between modules

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... some patient subgroups are **expected to respond similarly**.



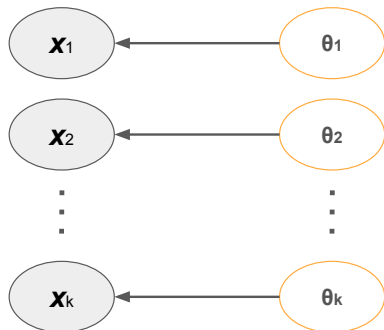
## Potential analysis strategies:

- Stand-alone analyses
- Complete pooling
- Borrowing of information

# Borrowing of information

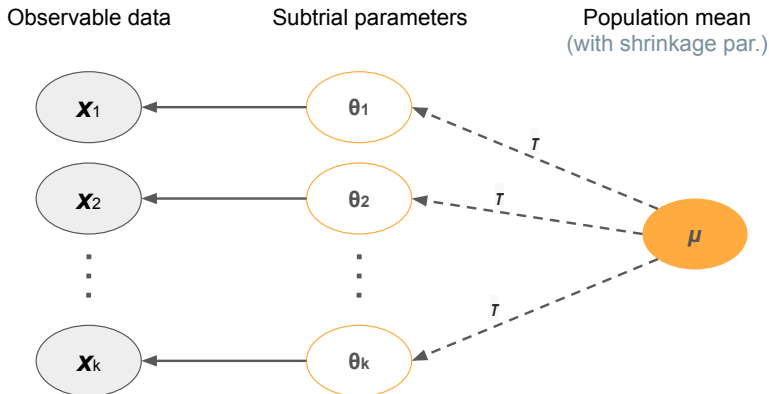
Observable data

Subtrial parameters



An independent analysis for NSCLC yields posterior estimate of 0.405 (see Lecture 1)

# Borrowing of information



An independent analysis for NSCLC yields posterior estimate of 0.405 (see Lecture 1)



The random variables  $\theta_1, \theta_2, \dots, \theta_K$  are exchangeable if

$$f_{\theta_1, \theta_2, \dots, \theta_K}(t_1, t_2, \dots, t_K) \stackrel{\text{distr.}}{=} f_{\theta_{\pi_1}, \theta_{\pi_2}, \dots, \theta_{\pi_K}}(t_1, t_2, \dots, t_K),$$

for any permutation  $(\pi_1, \dots, \pi_K)$  of the indices  $\{1, 2, \dots, K\}$ .

It can be shown that

- (1) i.i.d.  $\implies$  exchangeability,
- (2) exchangeability  $\implies$  identically distributed.

A Bayesian hierarchical model (BHM) for binomial data:

$$\begin{aligned} Y_i &\sim \text{Binomial}(n_i, p_i), & i = 1, \dots, k \\ \theta_i | \mu, \sigma &= \log \left( \frac{p_i}{1 - p_i} \right) \sim N(\mu, \sigma^2), \\ \mu &\sim N(\cdot, \cdot), \quad \sigma \sim g(\cdot). \end{aligned} \tag{1}$$

- ★ Hierarchical modelling assumes (similarity) of the  $\theta_i$ s
- Borrowing occurs between all baskets  $\rightarrow$  the estimates of the response rates are shrunk towards the common mean;
- Degree of shrinkage (i.e. borrowing) controlled by the so-called shrinkage/borrowing parameter,  $\sigma^2$ .
  - ▶  $\sigma^2 = 0 \rightarrow$  complete pooling of data from other baskets;
  - ▶  $\sigma^2 = \infty \rightarrow$  no borrowing.

# Choice of hyper-prior on borrowing parameter

Based on the uncertainty of  $\theta_i$  between different baskets, the following classification of values of  $\sigma$  was proposed

- **Small to moderate** heterogeneity:  $\sigma = 0.125$  to  $\sigma = 0.250$
- **Substantial to large** heterogeneity:  $\sigma = 0.5$  to  $\sigma = 1$

Various choices of hyper-prior  $g(\cdot)$  of  $\sigma$  were proposed with the most common options being

- **Inverse-Gamma**

Used in the original BHM proposal but was found to lead to a poor behaviour when  $\sigma^2$  is close to 0 (Cunanan et al. 2019)

- **Half-Cauchy** (with a moderately large scale was suggested instead), e.g. Half-Cauchy(0, 25) (Gelman 2006)
- **Half-Normal** with a prior standard deviation of  $s = 0.5$  [95% interval on  $\sigma$  is (0.02, 1.12)] or  $s = 1$  [95% interval on  $\sigma$  is (0.03, 2.24)] (Neuenschwander et al. 2016)

Now instead of analysing the first basket (8/20) alone, we conduct the analysis together with the second basket (6/18).

We use the BHM with Half-normal prior on  $\sigma^2$ .

The summary characteristics for the first basket:

|              | Independent   | BHM ( $s = 0.5$ ) | BHM ( $s = 1.0$ ) |
|--------------|---------------|-------------------|-------------------|
| Mean         | 0.405         | 0.380             | 0.385             |
| Median       | 0.402         | 0.377             | 0.381             |
| SD           | 0.104         | 0.087             | 0.091             |
| 95% CI       | (0.211,0.616) | (0.221,0.559)     | (0.218,0.574)     |
| Length of CI | 0.404         | 0.338             | 0.357             |

Assume that we were less lucky with the second basket and actually the number of responses was 0/18.

We again use the BHM with Half-normal prior on  $\sigma^2$ .

The summary characteristics for the first basket:

|              | Independent   | BHM ( $s = 0.5$ ) | BHM ( $s = 1.0$ ) |
|--------------|---------------|-------------------|-------------------|
| Mean         | 0.405         | 0.321             | 0.362             |
| Median       | 0.402         | 0.314             | 0.358             |
| SD           | 0.104         | 0.102             | 0.105             |
| 95% CI       | (0.211,0.616) | (0.145,0.536)     | (0.172,0.579)     |
| Length of CI | 0.404         | 0.391             | 0.407             |

- If the baskets are **truly homogeneous**, the gains are higher if smaller variance on the borrowing parameter is assumed
- However, if there is a **heterogeneous basket**, then borrowing can lead to too much shrinkage of the estimates and even increased variance of the estimate
- The choice of the distribution of the borrowing parameters  $\sigma^2$  should be based on the trade-off between these two cases.

# Relaxing the exchangeability assumption –

## EXNEX Neuenschwander et al. 2016

A Bayesian hierarchical model (BHM) assumes that the basket are exchangeable ... with probability of 1 (!).

The full exchangeability assumption is often violated. To tackle this, an EXNEX model was proposed

1. EX (exchangeable component): with prior probability  $w_i$ , basket  $i$  is exchangeable and a BHM is applied.
2. NEX (nonexchangeable component): with prior probability  $1 - w_i$ ,  $\theta_i$  is nonexchangeable with any other basket and analysed independently.

$$Y_i \sim \text{Binomial}(n_i, p_i),$$

$$\theta_i = \log \left( \frac{p_i}{1 - p_i} \right),$$

$$\theta_i = \delta_i M_{1i} + (1 - \delta_i) M_{2i},$$

$$\delta_i \sim \text{Bernoulli}(w_i),$$

$$M_{1i} \sim N(\mu, \sigma^2), \quad (\text{EX})$$

$$\mu \sim N(\cdot, \cdot),$$

$$\sigma \sim g(\cdot),$$

$$M_{2i} \sim N(m_i, \nu_i). \quad (\text{NEX}) \quad (2)$$

## Back to the example - I

Now instead of analysing the first basket (8/20) alone, we conduct the analysis together with the second basket (6/18).

We use the EXNEX with Half-normal prior on  $\sigma^2$  with  $s = 0.5$ .

The summary characteristics for the first basket:

|              | Independent   | BHM ( $s = 0.5$ ) | EXNEX ( $w = 0.5$ ) |
|--------------|---------------|-------------------|---------------------|
| Mean         | 0.405         | 0.380             | 0.395               |
| Median       | 0.402         | 0.377             | 0.390               |
| SD           | 0.104         | 0.087             | 0.097               |
| 95% CI       | (0.211,0.616) | (0.221,0.559)     | (0.216,0.597)       |
| Length of CI | 0.404         | 0.338             | 0.381               |



## Back to the example - II

Again, assume that we were less lucky with the second basket and actually the number of responses was 0/18.

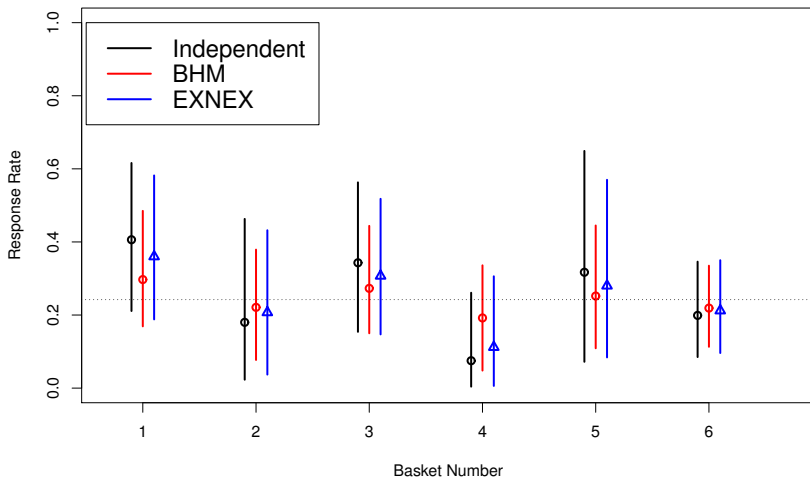
We again use the EXNEX with Half-normal prior on  $\sigma^2$  with  $s = 0.5$ .

The summary characteristics for the first basket:

|              | Independent   | BHM ( $s = 0.5$ ) | EXNEX ( $w = 0.5$ ) |
|--------------|---------------|-------------------|---------------------|
| Mean         | 0.405         | 0.321             | 0.402               |
| Median       | 0.402         | 0.314             | 0.399               |
| SD           | 0.104         | 0.102             | 0.105               |
| 95% CI       | (0.211,0.616) | (0.145,0.536)     | (0.204,0.614)       |
| Length of CI | 0.404         | 0.391             | 0.409               |

- EXNEX provides more flexibility: information is borrowed only between baskets assigned to the EX component but not from those in the NEX component;
- Weight  $w_i$  is the **prior** probability reflecting how likely we believe (a-priori) that the baskets are exchangeable
- It is uncommon to have strong prior information on the probability of exchangeability, so it is suggested to fix  $\pi_i = 0.5$
- This prior probability is updated to a some degree based on the homogeneity of the data but...
- The update might not be sensitive enough to the heterogeneity/homogeneity of responses in the typically small sample sizes

# Comparing different analysis models: VE-Basket



# Can we do anything better?

Some modules may display more similar treatment effects between themselves than with others.

Pertinent questions to the EXNEX approach:

- ★ How many EX distributions are needed?
- ★ What is an 'extreme subgroup'?



- ★ Construct a *complete* model space with all possible reclassification of subgroups with identical  $p_i$
- ★ Each model  $\mathcal{M}_\ell$  corresponds to a distinct model stipulation

Define  $\mathcal{M}_j$  as model  $j$  representing a permutation of basket allocation to the EX group or NEX group.

In the EX group, results are pooled and baskets have one shared response rate.

A weakly informative Beta prior is placed on the response rates, while a prior on each model,  $f(\mathcal{M}_j)$ , is also required. The posteriors  $f(p_i|\mathcal{M}_j)$  and  $f(\mathcal{M}_j|D)$  are computed.

BMA procedure to obtain a summary statistic for basket  $k$ . For example, for the probability of being above 0.25

$$\mathbb{P}(p_i > 0.25 | \text{Data, Prior}) = \sum_j \mathbb{P}(p_i > 0.25 | \mathcal{M}_j, D) f(\mathcal{M}_j | D)$$

- Many more models proposed
- Often single arm baskets considered, but randomized versions (borrowing on contrast parameter) available
- Often binary outcomes considered
- Basket structure in trials becoming more common, yet borrowing as main analysis still rare

- Is exchangeability clinically/mechanistically plausible
  - ▶ Any borrowing model will borrow if response is similar
- Is information in a basket dominated by borrowed information
- Control of false decision error



- Decision are made based on the **posterior probabilities for the treatment effect** given the clinical trial data;

For example, in a single-arm trial, one is claiming **benefit** if

$$\text{Prob}[\theta \geq p_0 \mid \text{Data, Prior}] > c$$

where

- $p_0$  is an effect threshold
- $c$  is probability threshold (often chosen to control type I error)
- “Data” is the observed trial data
- “Prior” is specified prior distribution on the treatment effect.

# Bayesian decision-making: example

The type I error is then defined as

$$\text{Prob}\{\text{Prob}[\theta \geq \rho_0 \mid \text{Data}, \text{Prior}] > c \mid \theta = \rho_0\}$$

The power is defined as

$$\text{Prob}\{\text{Prob}[\theta \geq \rho_0 \mid \text{Data}, \text{Prior}] > c \mid \theta = \rho_1\}$$

- What would be the null hypothesis in a basket trial setting?
- With  $K$  baskets, one can consider the configuration  $p_1 = p_2 = \dots = p_k = p_0$  where  $p_0$  is a null response rate for the calibration of the probability threshold  $c$ .
- This assumes that all (!) baskets are homogeneous. What would one expect to see if the some of the baskets has interesting treatment effect and some null?
- Were one to calibrate under the case of the maximum type I error, there will be no more advantages of using a basket trial design (Kopp-Schneider et al. 2020)

- Typically type I error controlled by tuning  $c$  using simulations under the global null
- RCaP
  - ▶ Defines several nulls of interest
  - ▶ Calibrate  $c$  such that type I error is controlled *on average* across selected null cases

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